

Week — Hypothesis testing I

Business Forecasting

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Where we are

C1 Framing → C2 Describing data → **C3 Uncertainty** → C4–C5 Regression → C6 Forecasting

We are in **C3 — uncertainty**. Last week we put a range around an estimate (a CI). This week we **test claims** about a parameter.

- **This week (Testing I): one-sample hypothesis testing** — hypotheses, test statistic, rejection region, Type 1 & 2 errors, power, and p-values.

Fast & hands-on: every test we set up, you run in R (`qnorm`, `pnorm`, `t.test`) — practice in `W06_learnr.Rmd`.

Hypothesis Testing

Hypothesis Testing

Can we get a higher price if cleanliness level is above 4.5?

This is a **hypothesis testing** question: we see which of two competing hypotheses is more likely given the data.

- **Hypothesis** is a claim about a parameter or a distribution.
- 1. **H_0 Null Hypothesis**: the claim to be tested, the skeptical perspective.
 - Prices of clean and dirty apartments are equal
 - $H_0 : \mu_c = \mu_d$
- 2. **H_A Alternative Hypothesis**: an alternative claim under consideration.
 - Prices of clean apartments are higher
 - $H_A : \mu_c > \mu_d$
- Assume the null is true and see if the data reject it in favor of the alternative.
- You either reject the null, or fail to reject it (never "accept" it).

Hypothesis Testing

Some examples of hypotheses:

- You released a promotion (10% off) to some customers. Do they spend a different amount than those who didn't get it?
 - $H_0 : \mu_p = \mu_n$ (promotion group spends the same)
 - $H_A : \mu_p \neq \mu_n$ (promotion group spends a different amount)
- A new snack gets an "excessive sugar" sticker if it exceeds 100g of sugar. You take a sample and test.
 - $H_0 : \mu_s = 100$ (average sugar is 100 or less)
 - $H_A : \mu_s > 100$ (average sugar is more than 100)

Hypothesis Testing

- Suppose we calculated the sample mean for sugar content. It's either below or above 100g.
- Why would we need a fancy test?
- Because we only have a sample, but we want to learn about the population parameter!
- Maybe in our sample it's below 100g, but in the whole population it's above.
- Hypotheses are always about parameters! Never about sample statistics.

Testing Procedure

A **test procedure** is a rule based on sample data for deciding whether to reject the null.

It relies on two elements:

- **Test statistic**
 - A function of the sample data that summarizes evidence for or against the hypothesis
 - We know how it is distributed under the null (e.g. standardized mean)
- **Rejection region**
 - Values of the test statistic for which we reject the null
 - These are values that are very unlikely under the null hypothesis
 - Typically threshold-based: reject H_0 if the test statistic $> k$

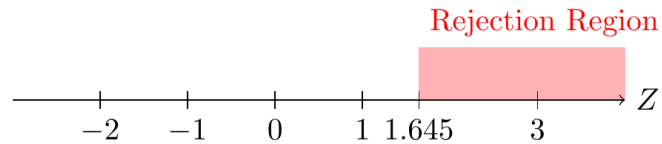
Testing Procedure: Example

Sugar example. Sample $n = 36$, population standard deviation known, $\sigma_x = 20$.

- Test statistic:

$$z = \frac{\bar{x} - 100}{\underbrace{\sigma_x / \sqrt{36}}_{SE}}$$

- Rejection region: for a 5% significance level, reject if $z \in \{1.645, \infty\}$



► `learnr W06_learnr.Rmd` → Exercise 3: Test statistic by hand

Testing Procedure Summary

1. Set up your hypotheses
2. Calculate your test statistic
3. Check it against the rejection region
4. Decision: Reject or fail to reject the null hypothesis

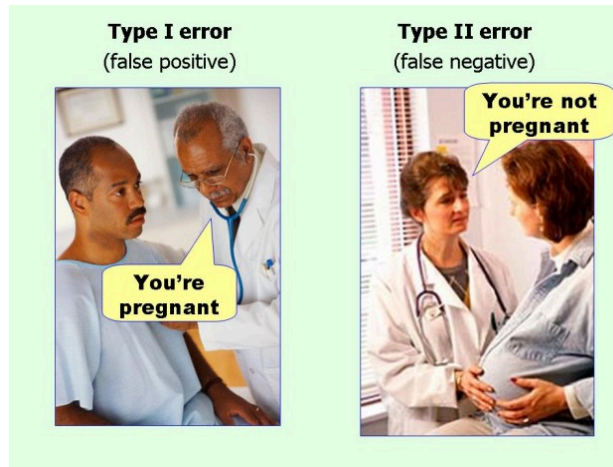
We will now go into the details of each step.

Decision errors

Decision errors

	Do not reject H_0	Reject H_0 in favor of H_A
H_0 true	Okay	Type 1 Error
H_A true	Type 2 Error	Okay

- H_0 : You are not pregnant
- H_1 : You are pregnant



Decision errors

Do people spend more if they get free shipping? (H_0 : they spend the same)

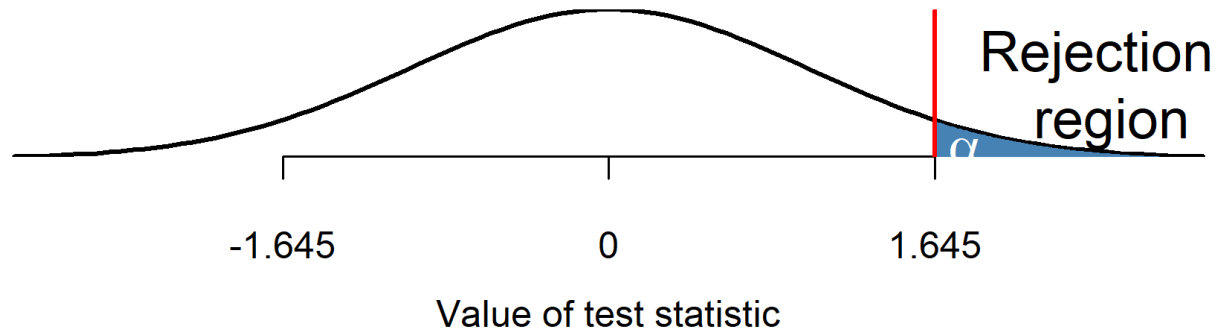
- **Type 1 error**: reject null when H_0 is true
 - Conclude free-shipping customers spend more, while in reality they spend the same
- **Type 2 error**: don't reject null when H_A is true
 - Conclude they spend the same, while free-shipping customers actually spend more
- Trial of a murderer: H_0 innocent, H_1 guilty. Describe error types 1 and 2.
 - Type 1: Convict an innocent
 - Type 2: Do not convict a murderer
- Probability of a type 1 error is α — the **size** or **significance level**.
 - We don't want to incorrectly reject the null more than $\alpha \cdot 100\%$ of the time.
- Probability of a type 2 error is β — to calculate it, you need the true μ .

Probability of type 1 error

- In the sugar example, suppose the null is true and $\mu_s = 100$.
- You take a sample of 36.
- Say you were unlucky: by chance you got rare high-sugar packages and $\bar{x} = 106$, so $z = \frac{106-100}{20/\sqrt{36}} = 1.8$. You reject the null — an error.
- Q: What is the probability the standardized mean of these 36 is in the rejection region (larger than 1.645)?
- What's the probability of making such an error?

$$\alpha = P_{H_0}(\bar{X} > 105.483) = P_{H_0}\left(\frac{\bar{X} - 100}{20/\sqrt{36}} > \frac{105.483 - 100}{20/\sqrt{36}}\right)$$

$$\stackrel{\text{CLT}}{=} P(Z > 1.645) = 1 - \Phi(1.645)$$



- Higher critical value = lower chance of mistake (lower α); lower critical value = higher α .
- Choosing the critical value determines α (and vice versa). We usually choose α , which fixes the rejection region.
- What is the critical value for $\alpha = 0.1$? For $\alpha = 0.01$? For $\alpha = 0$?

► [learnr W06_learnr.Rmd](#) → Exercise 8: What Type I error means

Probability of type 2 error

- What's the probability our sample mean is below the critical value, if the true sugar content is above 100?
- If I take 10000 samples from a distribution with true mean > 100 , what share of sample means fall below the rejection region?
- In our example:

$$\beta = P(\text{Type 2 error}) = \underbrace{P_{H_A}(z < 1.645)}_{\text{Pr. if alt is true}} = P_{H_A}\left(\frac{\bar{x} - 100}{20/\sqrt{36}} < 1.645\right)$$

- Problem: how is $\frac{\bar{x} - 100}{20/\sqrt{36}}$ distributed if the alternative is true?
- CLT does not apply to $\frac{\bar{x} - 100}{20/\sqrt{36}}$ because 100 is not the true mean under the alternative.
- We need to specify the true mean. Say $\mu_s = 110$. Then CLT applies to $\frac{\bar{x} - 110}{20/\sqrt{36}}$.

Probability of type 2 error

- Rewrite so we have $\frac{\bar{x}-110}{20/\sqrt{36}}$ on the left-hand side.

$$\beta = P_{H_A}\left(\frac{\bar{x} - 100}{20/\sqrt{36}} < 1.645\right) = P_{H_A}(\bar{X} < 105.483) = P\left(\frac{\bar{X} - 110}{20/\sqrt{36}} < \frac{105.483 - 110}{20/\sqrt{36}}\right)$$

- Now this becomes:

$$P\left(z < \frac{105.483 - 110}{20/\sqrt{36}}\right) = P(z < -1.3551)$$

- Where z is normally distributed thanks to CLT.

Probability of type 2 error

- More generally for this type of test:

$$\beta = P_{H_A}(\bar{X} < \mu_0 + z_{1-\alpha} \frac{\sigma_x}{\sqrt{n}}) = P\left(\frac{\bar{X} - \mu_s}{\sigma_x/\sqrt{n}} < \frac{\mu_0 - \mu_s}{\sigma_x/\sqrt{n}} + z_{1-\alpha}\right)$$

- Depends on the true μ and where the rejection region is.
- When you decrease the probability of a type 1 error, you increase the probability of a type 2 error.

► `learnr W06_learnr.Rmd` → Exercise 6: Decision at $\alpha = 0.05$

Trade-off between type 1 and 2 errors

$$\beta = P(\text{Type 2 error}) = P\left(Z < \frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right) = \underbrace{\Phi\left(\frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right)}_{\text{Normal CDF}}$$

$$\alpha = P(\text{Type 1 error}) = P(Z > z_{1-\alpha}) = 1 - \underbrace{\Phi(z_{1-\alpha})}_{\text{Normal CDF}}$$

If I change critical value $z_{1-\alpha}$, it has the opposite effect on them.

Interactive visualization: <https://shiny.rit.albany.edu/stat/betaprob/>

Power of a test

- Power is the probability of rejecting if the alternative is true.
- Power differs for every possible value of the alternative.
- Power for a one-sided test:

$$Power = P_{H_A}(Reject) = 1 - \underbrace{P_{H_A}(\text{Not Reject})}_{\text{type 2 error}} = 1 - \beta$$

$$Power = 1 - P\left(Z < \frac{\mu_0 - \mu_s}{\sigma_x / \sqrt{n}} + z_{1-\alpha}\right)$$

Power increases if: n increases; the alternative is further from the null; σ decreases; α increases.

A/B Test Power: Professional Photos

- Airbnb wants to test if professional photos increase weekly revenue.
- Design:
 - Control: no photos (baseline $\mu_0 = 2000$)
 - Treatment: professional photos (suspected $\mu_a = 2200$)
 - $\sigma = 600$, $n = 100$ per group, $\alpha = 0.01$
- Steps:
 1. State hypotheses: $H_0 : \mu_T - \mu_C = 0$ vs $H_A : \mu_T - \mu_C > 0$
 2. Find the critical value
 3. Compute power = $1 - \beta$
 4. Find minimum n for 80% power

LearnR: Power exercise — design the Airbnb photography A/B test, calculate power, and find minimum sample size

General Procedure

1. Determine the null and alternative hypotheses
2. Collect your sample of independent observations
3. Choose the appropriate test
4. Calculate the test statistic
5. Compare it to the rejection region
6. Reject or fail to reject the null hypothesis Let's talk about **appropriate tests!**

Choosing the Appropriate Test

The choice of test statistic and rejection region depends on:

- Type of data we are testing?
 - Single sample: $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$
 - Two independent samples: $H_0 : \mu_{s1} = \mu_{s2}$ vs $H_A : \mu_{s1} \neq \mu_{s2}$
 - Paired data: $H_0 : \mu_{d1} = \mu_{d2}$ vs $H_A : \mu_{d1} \neq \mu_{d2}$
- Can we estimate the standard deviation well?
 - Yes: large sample, or known standard deviation
 - No: small sample but normal distribution

Hypothesis Testing: Single Sample

Single Sample Test for Mean

General idea: test if the parameter equals/exceeds/falls below some value.

- Ex: $H_0 : \mu = 3$ vs $H_A : \mu \neq 3$
 - **Test statistic:** normalized sample mean

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{SE}$$

- SE depends on the estimate of the standard deviation:
 - Standard deviation known: $SE = \frac{\sigma}{\sqrt{n}}$
 - Standard deviation not known: $SE = \frac{s}{\sqrt{n}}$
- **Rejection region:** depends on the distribution
 - Large sample or known variance: $Z_{test} \sim N(0, 1)$ — critical values from standard normal
 - Small sample, normal data: $T_{test} \sim t(n - 1)$ — critical values from Student's t, $n - 1$ df

► learnr **W06_learnr.Rmd** → Exercise 2: Sample mean and standard deviation

Single Sample - Mean - Two Sided Test

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu \neq \mu_0$

Rejection Region: for a test of significance level α , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < z_{\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \mu_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- If small sample from normal

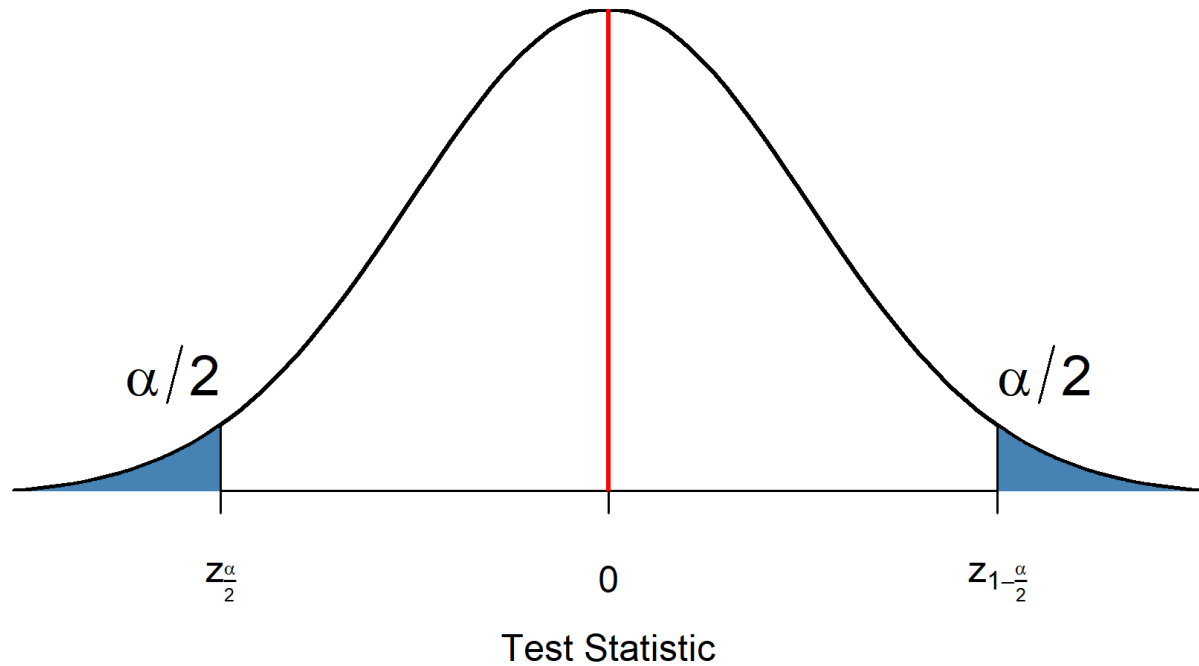
$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1), \frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X} - \mu_0}{SE} > t_{(n-1), 1-\frac{\alpha}{2}}$$

Where $z_{1-\frac{\alpha}{2}}$ and $t_{(n-1), 1-\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of the standard normal and Student's t with $n - 1$ df.

► `learnr W06_learnr.Rmd` → Exercise 4: Run the one-sample t-test

Single Sample - Mean - Two Sided Test

Distribution under H_0



Link between Tests and Confidence Intervals

Test for μ with $H_0 : \mu = 4$ and $H_A : \mu \neq 4$.

- Population standard deviation known, $\sigma = 3$. Sample of 49 with mean $\bar{x} = 3$.
- Test statistic: $z_{test} = \frac{3-4}{\sigma/\sqrt{49}} = \frac{3-4}{3/7} = -2.34$
- Critical values at $\alpha = 0.05$ are -1.96 and 1.96 , hence we reject.
- The 95% confidence interval for the mean:

$$\left\{3 - 1.96\frac{\sigma}{\sqrt{49}}, 3 + 1.96\frac{\sigma}{\sqrt{49}}\right\} = \{2.16, 3.84\}$$

- It does not contain the null hypothesis.

Link between two sided Tests and Confidence Intervals

- More generally: the $1 - \alpha$ CI doesn't contain the null $\iff$ the null would be rejected with a test of significance α .

https://kzaremba.shinyapps.io/Hypothesis_Confidence/

- Mathematically: let μ_0 be the null. In a two-sided test we reject at α if

$$\left| \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \right| > z_{1-\frac{\alpha}{2}}$$

- Which can be rewritten as:

$$\mu_0 < \bar{X} - z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \quad \text{or} \quad \mu_0 > \bar{X} + z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

LearnR: CI-test equivalence, and a one-sample t-test for a Twitter campaign (≥ 200 likes/tweet)

Single Sample - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu < \mu_0$

- This includes any null like $\mu \geq \mu_0$. Why?
- If you rejected μ_0 at α , you would reject for sure anything larger than μ_0 . **Rejection Region:** for a test of significance level α , reject
- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} < z_\alpha$$

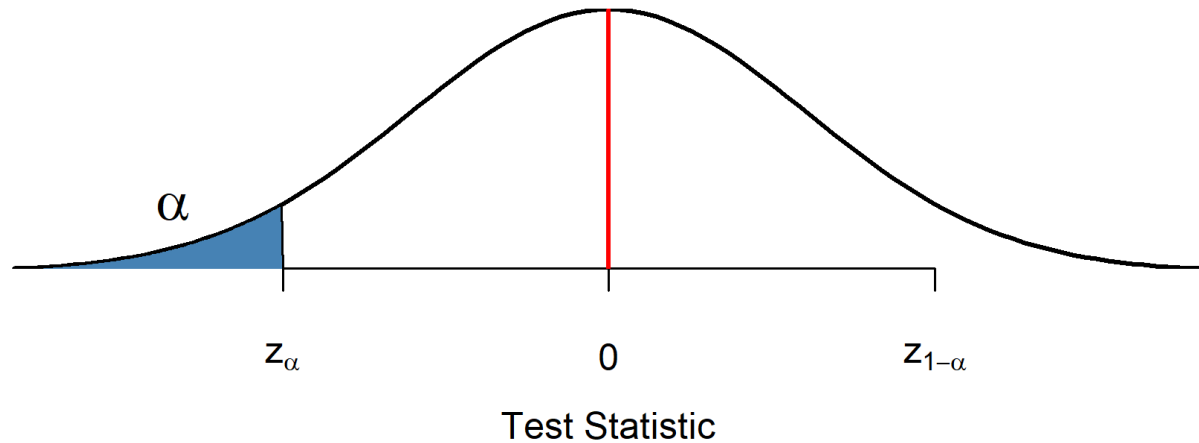
- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1),\alpha}$$

Where z_α and $t_{(n-1),\alpha}$ are α quantiles of the standard normal and Student's t with $n - 1$ df.

Single Sample - Mean - One sided - Case 1

Distribution under H0



- **Intuition:** never reject if $\bar{X} \geq \mu_0$; reject if $\bar{X}$ is sufficiently smaller than μ_0 . If the null is true ($\mu = \mu_0$) we reject with probability

$$P\left(\frac{\bar{X} - \mu_0}{s/\sqrt{n}} < z_\alpha\right) = P(Z < z_\alpha) = \alpha$$

Single Sample - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu > \mu_0$

- This includes any null like $\mu \leq \mu_0$. Why?
- If you rejected μ_0 at α , you would reject for sure anything smaller than μ_0 . **Rejection Region:** for a test of significance level α , reject
- If large sample or known variance from normal

$$\frac{\bar{X} - \mu_0}{SE} > z_{1-\alpha}$$

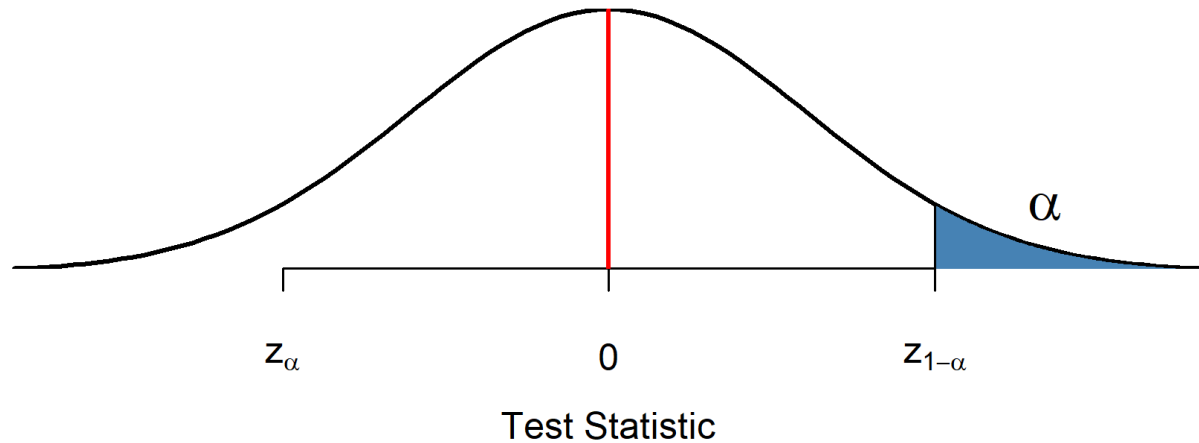
- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} > t_{(n-1), 1-\alpha}$$

Where $z_{1-\alpha}$ and $t_{(n-1), 1-\alpha}$ are $(1 - \alpha)$ quantiles of the standard normal and Student's t with $n - 1$ df.

Single Sample - Mean - One sided - Case 2

Distribution under H0



- **Intuition:** never reject if $\bar{X} \leq \mu_0$; reject if $\bar{X}$ is sufficiently larger than μ_0 . If the null is true ($\mu = \mu_0$) we reject with probability

$$P\left(\frac{\bar{X} - \mu_0}{s/\sqrt{n}} > z_{1-\alpha}\right) = P(Z > z_{1-\alpha}) = \alpha$$

► learnr W06_learnr.Rmd → Exercise 7: One-sided vs two-sided

EXAM QUESTION — Fall 2025 Midterm 1

Real Midterm 1 questions — Households & Environment Survey ($n = 13,255$). We solve these in class using `Data/households_environment.rda` (object `exam_data`).

Q9. A report claims the true mean monthly electricity bill is 200. Your sample mean is 206. Why can't we immediately conclude the report is wrong?

Q10. State (H_0) and (H_A) for testing whether the true mean monthly electricity bill differs from 200.

A sample mean of 206 could arise by chance even if $\mu = 200$ — sampling variability. We need a **test** (Q10: $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$).

► `learnr W06_learnr.Rmd` → Exercise 1: State the hypotheses

P-values

Definition: the p-value is the probability, under the null hypothesis, of obtaining a test statistic at least as extreme as the one computed from the sample.

Depends on: your hypothesis, and the calculated test statistic.

- Suppose under H_0 the test statistic is $Z_{test} \sim N(0, 1)$.
- One-sided tests:
 - $H_A : \mu > \mu_0$: $p\text{-value} = P(Z_{Test} \geq \text{computed statistic})$
 - $H_A : \mu < \mu_0$: $p\text{-value} = P(Z_{Test} \leq \text{computed statistic})$
- Two-sided test:
 - $H_A : \mu \neq \mu_0$: $p\text{-value} = 2P(Z_{Test} \geq |\text{computed statistic}|)$

P-values: Example

- Test $H_0 : \mu = 5$ vs $H_A : \mu > 5$.
- In a sample of 36 we have $\bar{x} = 5.5$ and $s = 2$.
- The test statistic is distributed normally:

$$p\text{-value} = P(Z > \frac{5.5 - 5}{2/\sqrt{6}}) = P(Z > 1.5) = 0.066$$

Intuitively: the probability that a statistic we calculated (or a more extreme value) could arise just by chance if the null is true.

P-value in two-sided test

- In a two-sided test, the p-value accounts for **both tails**.
- We double the one-tail probability: **p-value** = $2P(Z \geq |z_{test}|)$.

Interactive visualization: <https://rpsychologist.com/pvalue/>

► learnr W06_learnr.Rmd → Exercise 5: Read the p-value

P-values and Test Significance

- In our example we calculated the p-value of 0.066.
- Would a test at $\alpha = 0.05$ reject the null?
- Any test with significance level $\alpha > p\text{-value}$ would reject the null.

Another way to think about p-values:

- The p-value is the smallest significance level α at which H_0 can be rejected.

(3) The average grade among fourth graders on a math aptitude test in a certain state is 62.5 out of 100 points. (Assume 62.5 is the population average.) An educational company has developed an app for elementary-school math and would like to test whether the app improves average performance on the aptitude test among fourth graders. Let μ_X denote the population mean of test scores among students who use the app.

- (a) To provide statistical evidence that the app *increases* the true average of test scores, what are the appropriate one-sided null hypothesis and alternative hypothesis?
- (b) The company provides the app to 15 randomly selected students. The sample mean and sample standard deviation of test scores are 68.3 and 13.5, respectively. Under the assumption that the test scores are i.i.d. and normally distributed, would the *t*-test of the null hypothesis in (a) be rejected at the 5% level? at the 10% level?
- (c) What is the p-value associated with the *t*-test in (b)?

EXAM QUESTION — Fall 2025 Midterm 1

Q12. Using the electricity-bill data ($n = 13,255$), compute the test statistic for $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$.

We solve this in class on `Data/households_environment.rda` (object `exam_data`):

- Compute $\bar{x}$ and s , then $z = \frac{\bar{x} - 200}{s/\sqrt{n}}$ with $n = 13,255$.
- Large $n \rightarrow$ compare to the **normal** critical values, or read off the **p-value**.

Practice now: the one-sample z/t problems in the Week (Testing I) set, then re-run Q12 on `exam_data`.

What you must be able to do this week

- State H_0 and H_A about a parameter (never a sample statistic), one- or two-sided.
- Build the test statistic $\frac{\bar{X} - \mu_0}{SE}$ and pick the right reference distribution (normal vs Student's t).
- Define the **rejection region** for a chosen α and reach a **decision**.
- Explain **Type 1 vs Type 2 errors**, the significance level α , β , and **power** — and their trade-off.
- Compute and interpret a **p-value**, and use the **test $\Leftrightarrow$ confidence interval** equivalence.

Next week: comparing **two** groups — difference in means, paired data, correlation, and A/B testing.

Appendix — optional (extra detail)

Testing Procedure: Example

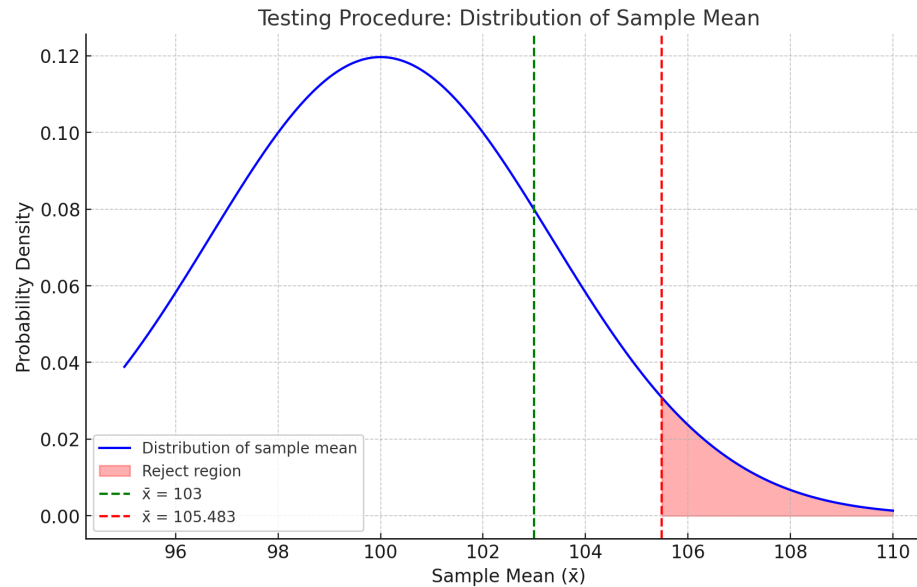
- Suppose I reject if $z > 1.645$, with $\sigma_x = 20$. How large must the mean be to reject?

Reject if:

- $\frac{\bar{x} - 100}{20/\sqrt{36}} > 1.645$
- Or equivalently if $\bar{x} > 105.483$
- Those two formulations are equivalent.
- Why don't we reject even if $\bar{x} = 103$?

Testing Procedure: Example

- Because there is randomness in the sample:
- Even if $\mu_s = 100$, we could get a sample with $\bar{x} > 100$.



- Why $\bar{x} > 105.483$ though? Formally later, but it's very unlikely to get a sample mean that large if $\mu_s = 100$.